

# Talha Ahmed

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## EDUCATION & RESEARCH BACKGROUND

**Sungkyunkwan University (SKKU)** Sep. 2026 – Present  
*M.S. Computer Science and Engineering – DASH Lab (Advisor: Prof. Simon S. Woo)* Suwon, South Korea

**Lahore University of Management and Sciences (LUMS)** Sep. 2020 – May 2024  
*B.S Mathematics - Economics (Joint Major) + Minor in Computer Science* CGPA/Minor GPA: 3.83/3.87

**Relevant Courses:** Real Analysis, Adv. Calculus, Applied Probability, Data Structures, Machine Learning, Convex Optimization, Data Mining, Deep Learning, Adv. Signal Processing, Reinforcement Learning, Adv. Econometrics, Generative AI, Numerical Analysis

**Research Interest Keywords:** Deepfake Detection, AI Safety & Security, Generative Models (Diffusion/Score-Based), Computer Vision, Computational/Medical Imaging, Inverse Problems, Representation Learning. See [Presentation Overview](#)

## RESEARCH EXPERIENCE

**Graduate Researcher (M.S.)** Fall 2026 – Present  
*Prof. Simon S. Woo – DASH Lab @ SKKU* Suwon, South Korea

- Research at the intersection of computer vision, generative models, and AI safety, with a primary focus on **deepfake detection** and the robustness of vision systems to manipulation and adversarial attacks.

**Research Assistant** | Pytorch, MONAI, Diffusers, HuggingFace, OpenCV | Summer 2024 – Summer 2026  
*Dr. Hassan Mohy-ud-Din – Algorithms in Theory & Practice Lab @ LUMS* Lahore, Pakistan

- Research centered on generative models and their applications in medical imaging and inverse problems, including MRI reconstruction, binary/multi-class segmentation, image restoration, and super-resolution.
- First author of **Wave-GMS** (ICASSP 2026): lightweight multi-resolution generative model (~2.6M params) achieving state-of-the-art segmentation and cross-domain generalization on medical imaging datasets.
- Compiled notes on mathematical inequalities, e.g. [Jensen's Inequality](#), and their applications in data science and information theory.

**Senior Year Thesis** | Pytorch, Numpy, MATLAB, Spatio-Temporal Analysis | Summer 2023 – Summer 2024  
*Dr. Muhammad Tahir* Lahore, Pakistan

- Senior year project on **Model-Based Deep Learning** for matrix completion. Proposed ConvMC-Net, a convolutional network replacing nuclear-norm updates with trainable layers, achieving faster and more accurate recovery compared to ALM and ADMM-Net.
- Extended this work with ConvHuberMC-Net, introducing a convex Huber criterion and unfolded Majorization–Minimization framework to improve robustness to impulsive GMM noise, eliminate explicit singular value thresholding, and enable parallelizable computations. See [Report + Presentation](#).

**Directed Research Project** | R, Shiny, HTML, CSS | Spring 2023  
*Dr. Ihsan Ayub Qazi – Network Systems Group @ LUMS* Lahore, Pakistan

- Developed a *Shiny* app to predict digital literacy using a pre-trained Random Forest model. See [GitHub](#).

## PUBLICATIONS

### Wave-GMS: Lightweight Multi-Scale Generative Model for Medical Image Segmentation

Talha Ahmed, Nehal Ahmed Shaikh, and Hassan Mohy-ud-Din. *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) 2026*. See [IEEE Xplore](#) | [Github](#) | [arXiv](#).

### Unified Perspective on Diffusion Models: Theory and Practice in Medical Imaging and Inverse Problems

Talha Ahmed and Nehal Ahmed Shaikh. *Survey manuscript in preparation*. See [Draft](#).

## TEACHING + WORK EXPERIENCE

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### EE 563/MATH 325: Convex Optimization (Spring 2025)

Professor Hassan Mohy-ud-Din

Teaching Assistant

- Held weekly office hours, made and graded assignments, and engaged in semi-formal student counseling

### ACTA 6304: Advanced Machine Learning (Fall 2024)

Professor Momin Ayub Uppal

Teaching Assistant

- Held weekly office hours, made and graded assignments, and engaged in semi-formal student counseling

### CS 535: Machine Learning (Spring 2024)

Professor Momin Ayub Uppal

Teaching Assistant

- Held weekly office hours, invigilated quizzes and exams, held tutorials, made and graded assignments, and engaged in semi-formal student counseling

### EDUX 562: Data Lab (Spring 2023)

Professor Ahmad Ayub

Teaching Assistant

- Held weekly office hours, invigilated STATA labs, graded assignments, and engaged in semi-formal student counseling

### ECON 221: Intermediate Macroeconomics (Fall 2022)

Professor Usman Elahi

Teaching Assistant

- Held weekly office hours, conducted assignment tutorials, created/reviewed/invigilated/graded quizzes, created/reviewed/solved assignments, and engaged in semi-formal student counseling

### STATA Workshop (Dec 2022 - Jan 2023)

Professor Usman Elahi

Teaching Assistant

- Assistant to Professor Usman Elahi for 'Capacity Building and Training on Data Management & Analysis Using STATA', organized in collaboration with the Bureau of Statistics, Government of Punjab, for Statistical Officers.

## UNDERGRADUATE COURSE PROJECTS/PRESENTATIONS

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### Speech Recognition and Translation System | Chroma DB, Langchain, LlamaIndex, Unsloth, Gradio | Spring 2024

CS 5302: Generative AI for Natural Language and Speech Processing

- We aimed to develop an application that can interpret, translate, and vocalize spoken language in real-time, and is specifically catered for patient-doctor conversations.
- We integrated various open source models of Automatic Speech Recognition, Neural Machine Translation, and Text-to-Speech synthesis. See [Project Deliverables](#) | [Github](#).

### Reinforcement Learning Algorithms on Tic-Tac-Toe | Gymnasium, Pytorch, Numpy |

Fall 2023

CS 6314: Dynamic Programming and Reinforcement Learning

- Trained a reinforcement learning agent to play 2D and 3D Tic-Tac-Toe using algorithms like Value Iteration, Temporal Difference Learning, and Deep Q Networks. See [Project Report](#) | [Github](#).

### Panel Data and Tobit Analysis on Health Care Dataset | STATA, Hypothesis Testing, Tobit |

Fall 2023

ECON 438: Econometrics II

- Conducted panel data and Tobit analysis on a German healthcare dataset to determine factors influencing doctor or hospital visits using fixed/random effects and tobit models. See [Project Report + Source Code](#).

### Clustering, Association and Frequent Pattern Mining | Scikit-learn, Numpy, Pandas, Mlxtend |

Spring 2023

CS 432: Introduction to Data Mining

- Analyzed drug consumption patterns in Connecticut, USA using DBSCAN, Apriori, and Fpgrowth algorithms for clustering, association, and frequent pattern mining. See [Project Report](#).

### Sentiment Analysis on Audio Recordings | Scikit-learn, Numpy, Librosa, Hyperparameter Tuning |

Spring 2023

CS 535: Machine Learning

- Identification and extraction of features followed by a mathematical background of some popular machine learning methods and their performance evaluation. See [Project Report](#).

## HONORS & ACADEMIC DISTINCTIONS

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- Awarded the **Seoul Tech Scholarship (STS) 2026** for graduate study at Sungkyunkwan University (SKKU)
- Ranked in the **top 9%** of LUMS SBASSE Batch (300+ students) of 2024
- Placed on **Dean's Honor List (CGPA  $\geq$  3.60)** for **2020-2021, 2021-2022, 2022-2023**
- Graduated with **Dean's Honour List** and **High Distinction (CGPA  $\geq$  3.80)**

## TECHNICAL SKILLS

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**Languages:** C++, Python, STATA, MATLAB, R, HTML/CSS

**Programming Frameworks:** PyTorch, OpenCV, Shiny, Numpy, Pandas, HuggingFace, MONAI, Diffusers, Accelerate

**Tools:** Linux, Git, Dropbox, L<sup>A</sup>T<sub>E</sub>X, Microsoft, VS Code, Google Colab, Jupyter Lab